APSTA-GE 2044

Generalized Linear Models: Lecture 5

Klint Kanopka

New York University

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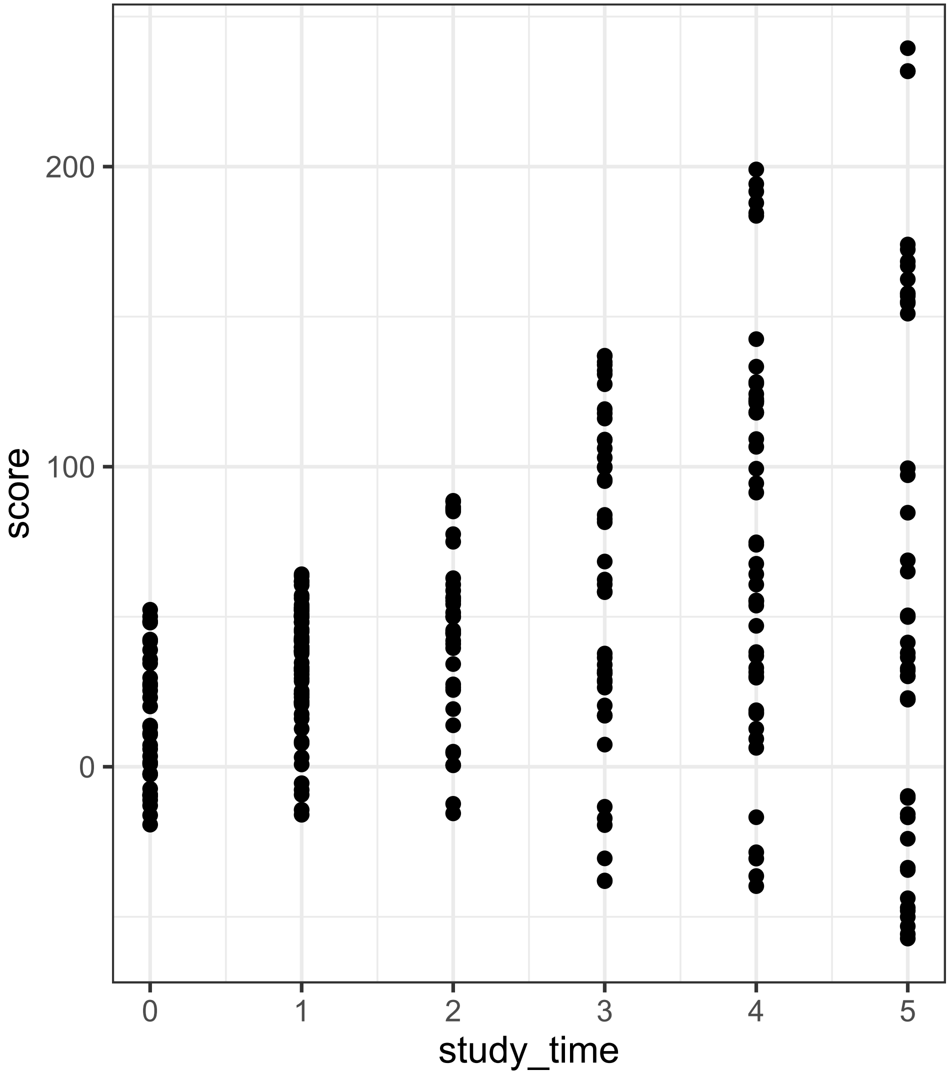
The Setup

- We have 9 classrooms, each with a different teacher
- Each teacher has 30 students
- Students were given a math test
- We got their scores!
- We also got how many hours each student reported studying for the test!
- Let’s take a look…

Looking at the data

```r
ggplot(d, aes(x = study_time, y = score)) +
  geom_point() +
  guides(color = 'none') +
  theme_bw()
```

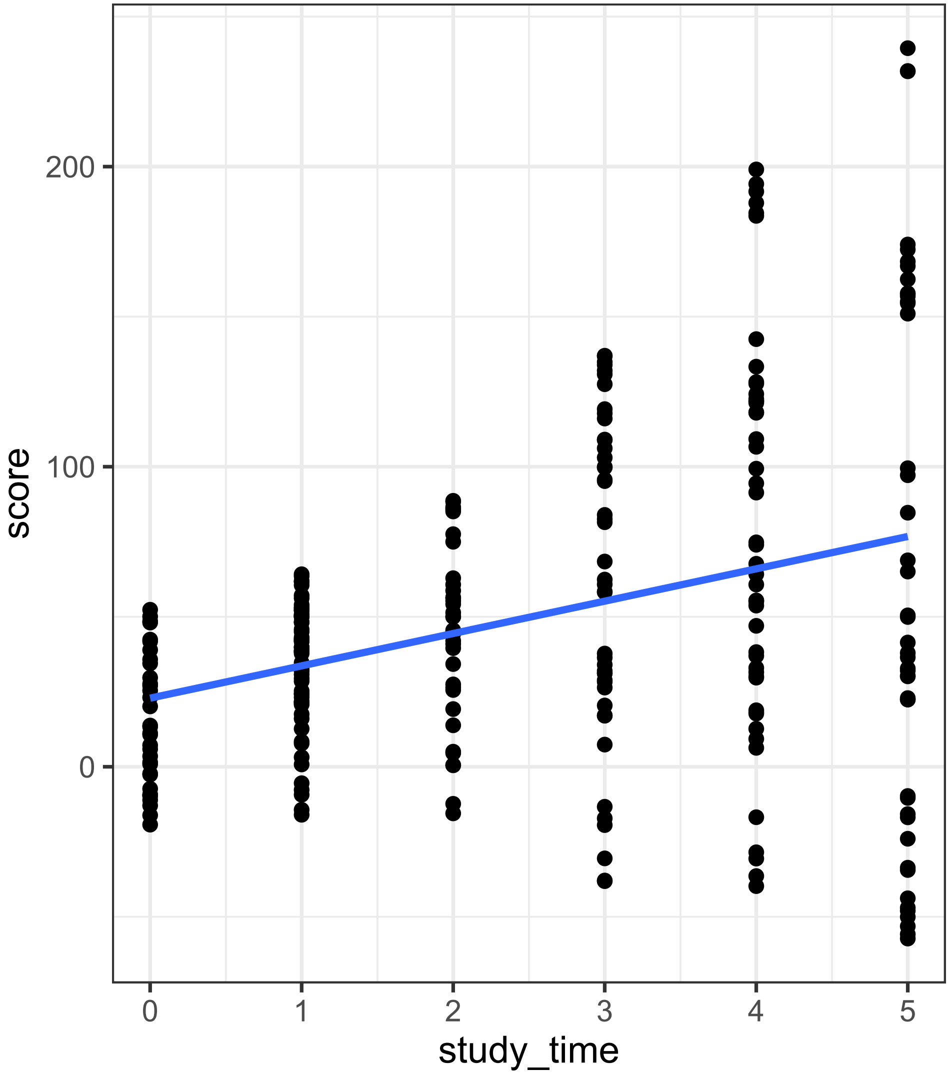
- What do you see happening?
- Is studying associated with better scores?



Looking at the data

```r
ggplot(d, aes(x = study_time, y = score)) +
  geom_point() +
  geom_smooth(method = 'lm', se = F) +
  guides(color = 'none') +
  theme_bw()
```

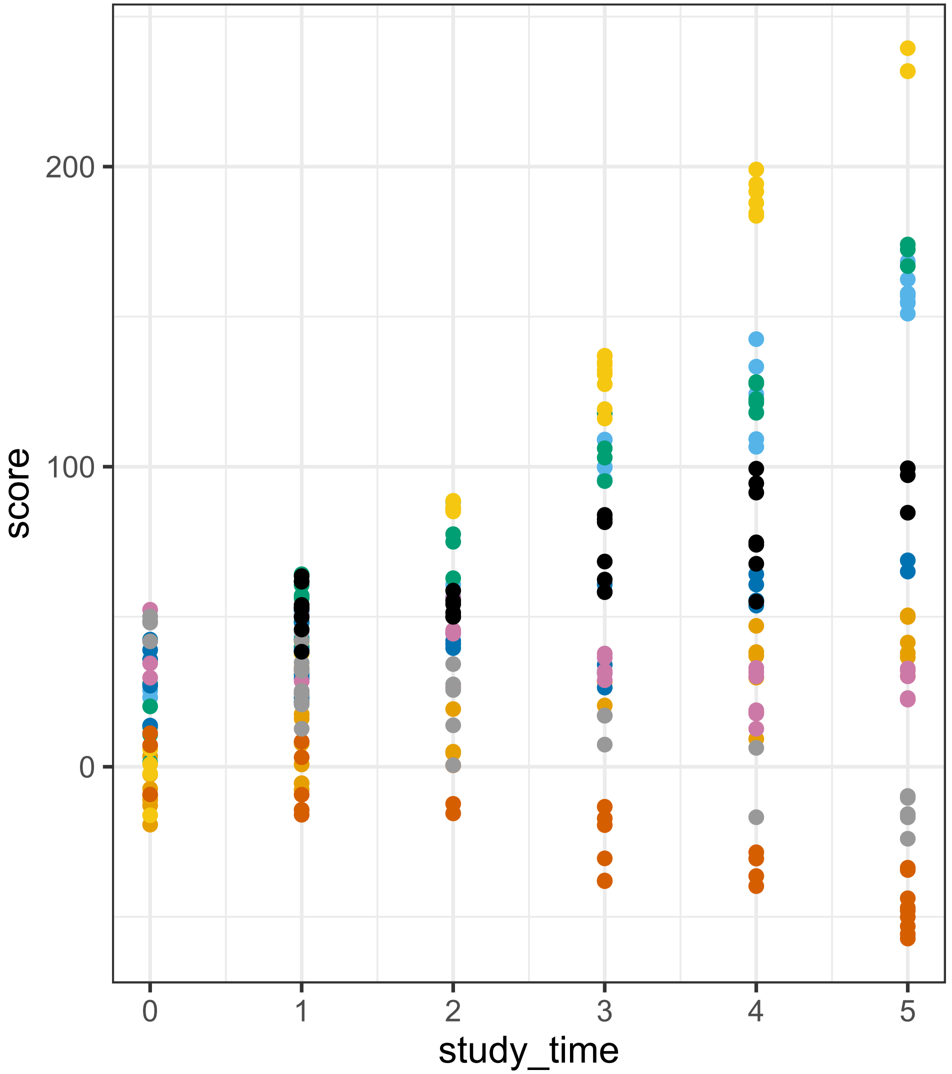
- On average, studying helps!



Does classroom assignment matter?

```r
ggplot(d, aes(x = study_time, y = score,
         color = as.character(classroom))) +
  geom_point() +
  scale_color_okabeito() +
  guides(color = 'none') +
  theme_bw()
```

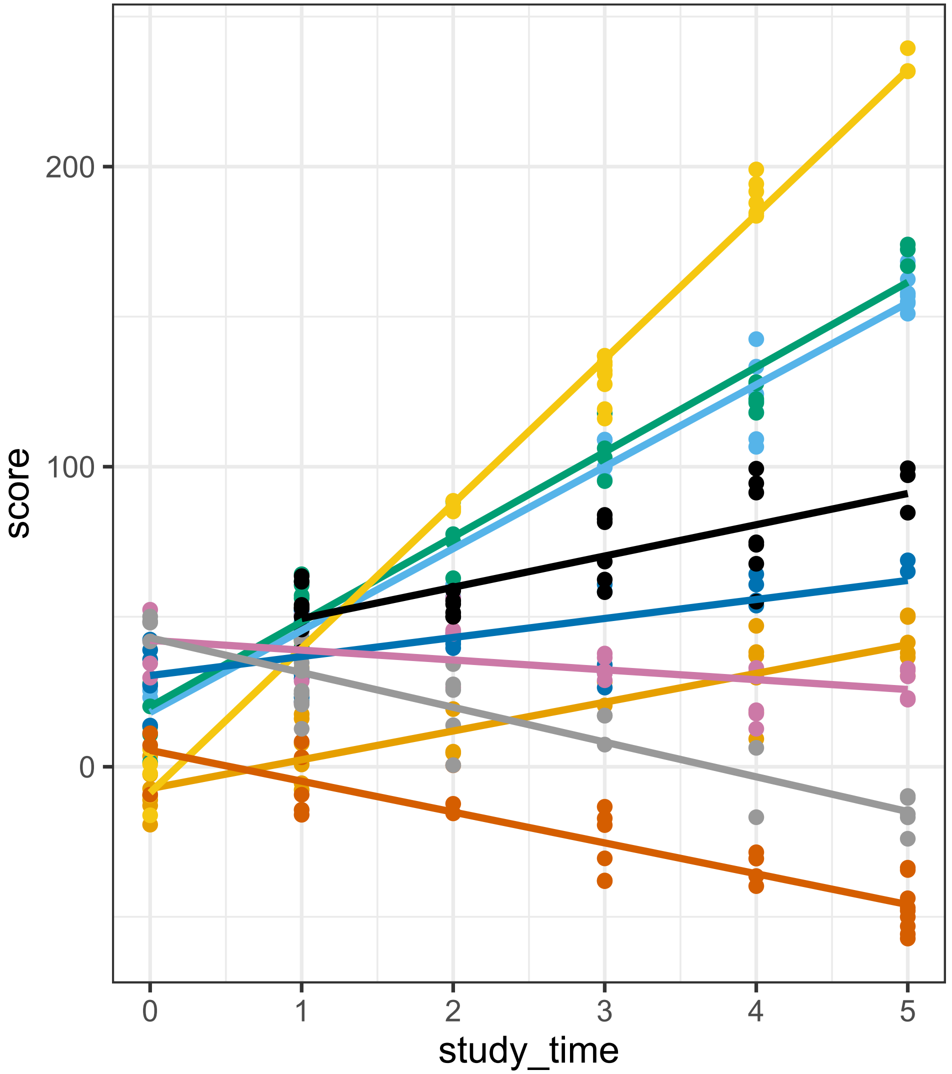
- What about variation between classrooms?
- Does the grouping make a difference?



Does classroom assignment matter?

```r
ggplot(d, aes(x=study_time, y=score,
        color=as.character(classroom))) +
  geom_point() +
  geom_smooth(method='lm', se=F) +
  scale_color_okabeito() +
  guides(color='none') +
  theme_bw()
```

- It seems that classroom matters a lot!
- How can you model this?
  - Consider a version that uses only *fixed effects*
  - Consider a version that uses *random effects*
  - How many parameters does each estimate?



Simulating Multilevel Data

- What does a data generating process look like for something like this?

```r
set.seed(242424)

# simulate data for individuals, assign to groups
d_sim <- data.frame(student_id = 1:(9*30))
d_sim$classroom <- rep(1:9, each=30)
d_sim$study_time <- sample(0:5, 9*30, replace=T)

# make a dataframe for group characteristics
d_classroom <- data.frame(classroom = 1:9)
d_classroom$teacher_effect <- rnorm(9, mean = 30, sd = 20)
d_classroom$teacher_factor <- rnorm(9, 1, 2)

# join them together - left join group to individuals
d_sim <- left_join(d_sim, d_classroom, by = 'classroom')

# generate your outcome variable, don't forget to add noise!
d_sim$score <- d_sim$teacher_effect + 10*d_sim$teacher_factor*d_sim$study_time + rnorm(nrow(d), mean = 0, sd = 10)
```

Estimating MLMs

Run the code from the previous slide to get the simulated data, then for each of the following models:

1. What is it estimating?
2. Fit the model in `R`
3. Look at the `summary()`
4. How do you interpret the parameters?

```
lm(score ~ study_time, data = d_sim)

lm(score ~ study_time + as.character(classroom), data = d_sim)

lmer(score ~ study_time + (1|classroom) + study_time, data = d_sim)

lm(score ~ as.character(classroom)*study_time, data=d_sim)

lmer(score ~ study_time + (1 + study_time | classroom), data = d_sim)
```

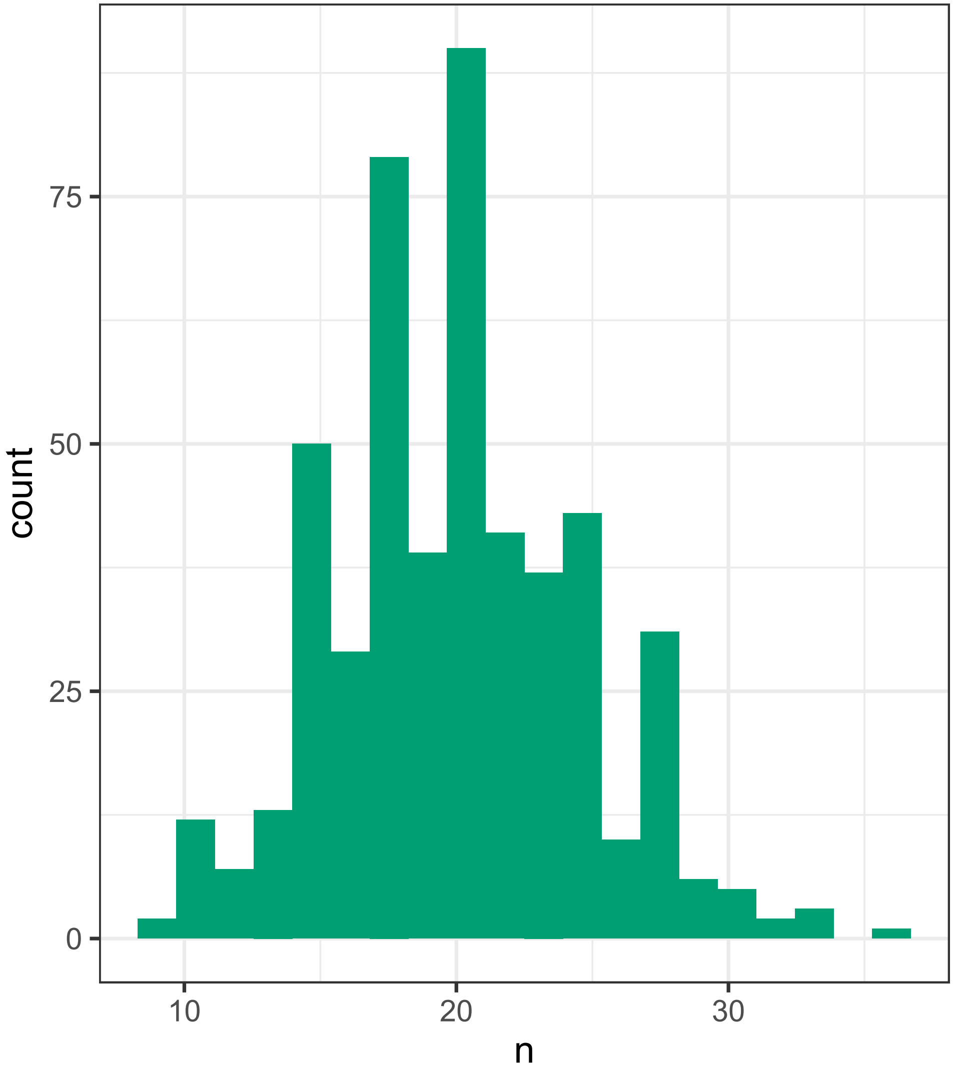
Working with Multilevel Data

Exploring the Data

- Download `lect-05-data.csv`
- First:
  1. How many students are there?
  2. How many teachers are there?
  3. How many schools are there?
  4. Visualize the distribution of students per teacher
  5. Visualize the distribution of teachers per school
  6. Visualize the distribution of students per school
- For each variable (except the `_id` variables):
  1. Determine if it is measured at the student, teacher, or school level
  2. Visualize the distribution

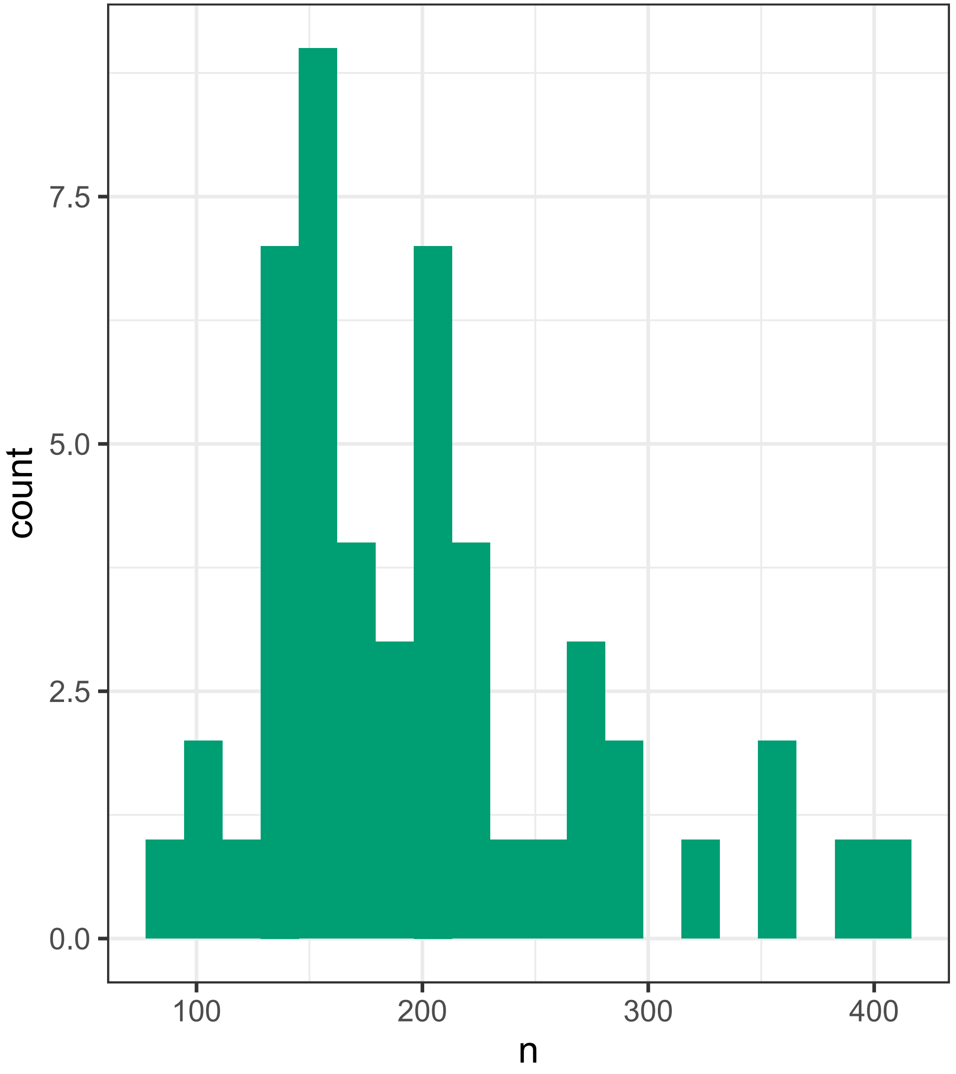
Students per Teacher

```r
d |>
  count(teacher_id) |>
  ggplot(aes(x = n)) +
  geom_histogram(
    bins = 20,
    fill = okabeito_colors(3)) +
  theme_bw()
```



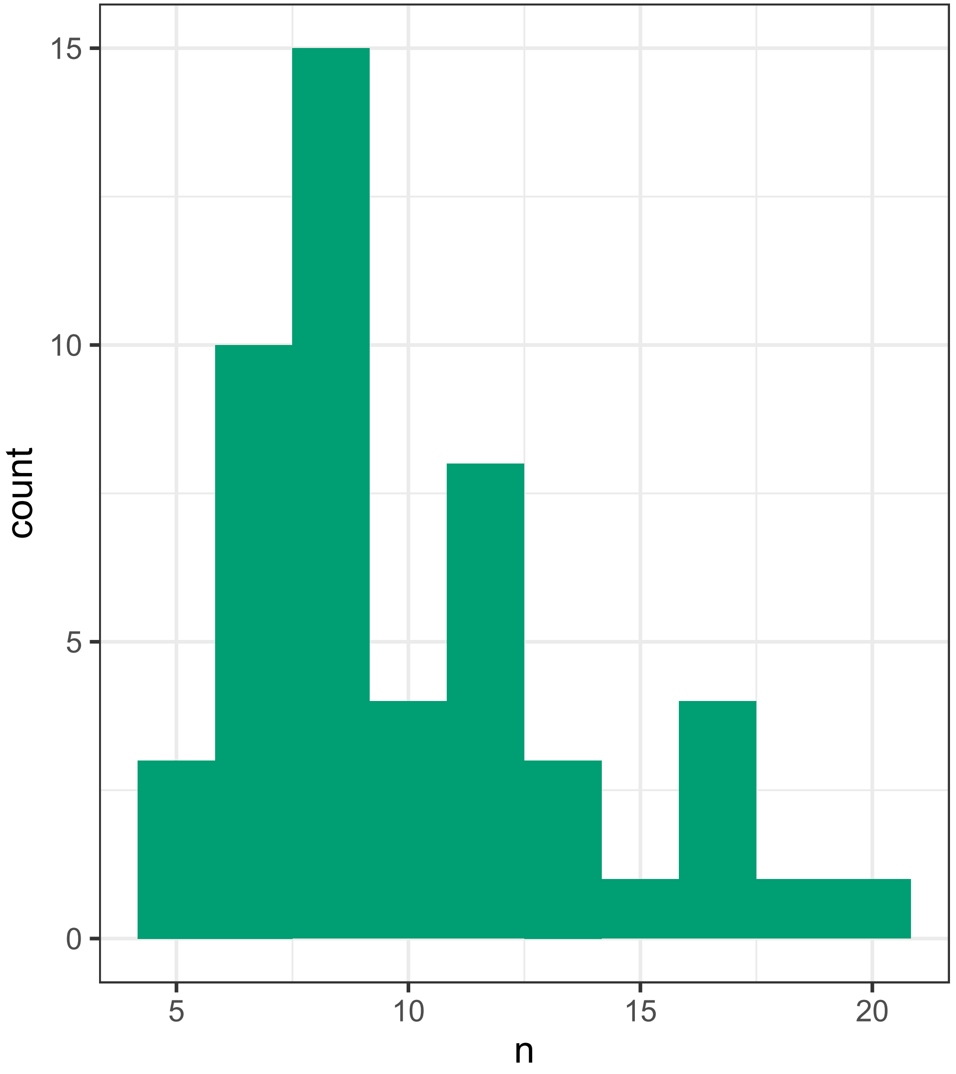
Teachers per School

```r
d |>
  count(school_id) |>
  ggplot(aes(x = n)) +
  geom_histogram(
    bins = 20,
    fill = okabeito_colors(3)) +
  theme_bw()
```



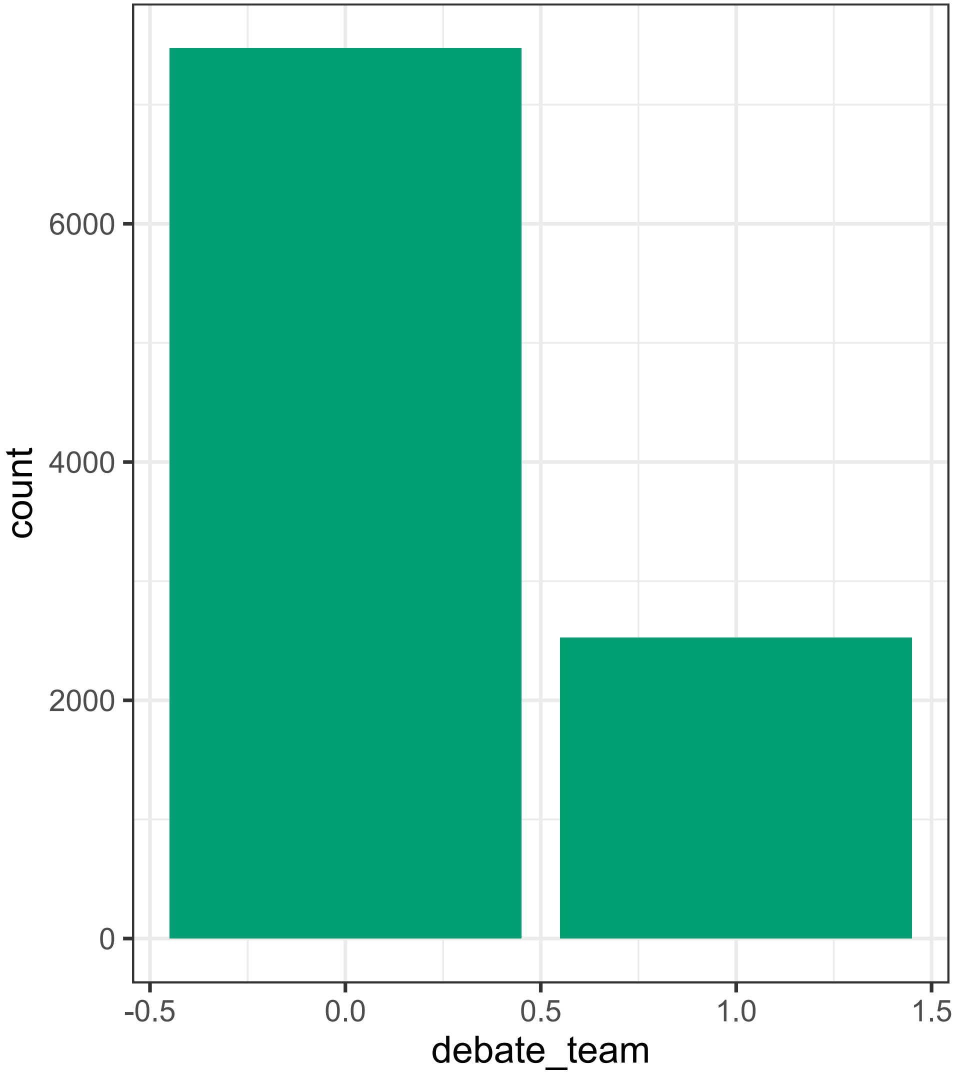
Students per School

```
d |>
  select(teacher_id, school_id) |>
  distinct() |>
  count(school_id) |>
  ggplot(aes(x = n)) +
  geom_histogram(
    bins = 10,
    fill = okabeito_colors(3)) +
  theme_bw()
```



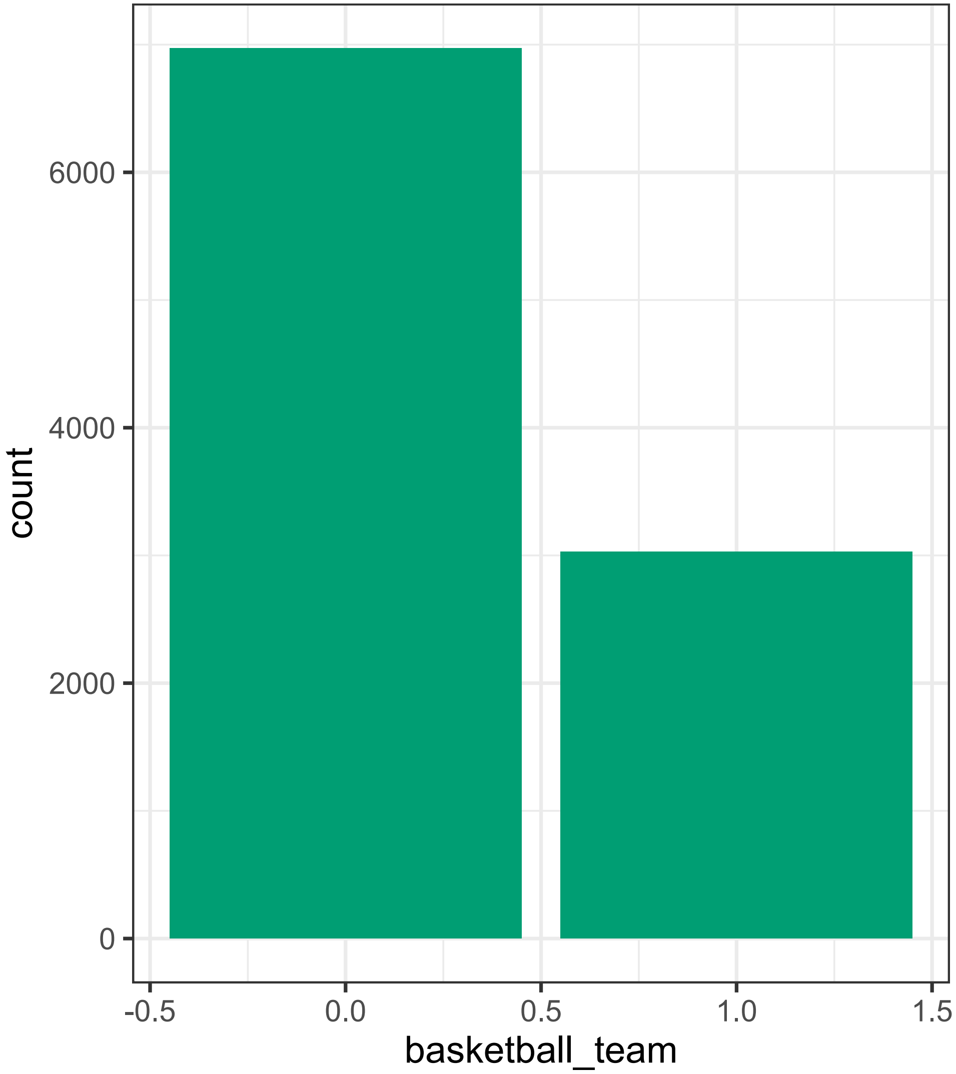
Debate Team

```r
ggplot(d, aes(x = debate_team)) +
 geom_bar(fill = okabeito_colors(3)) +
 theme_bw()
```



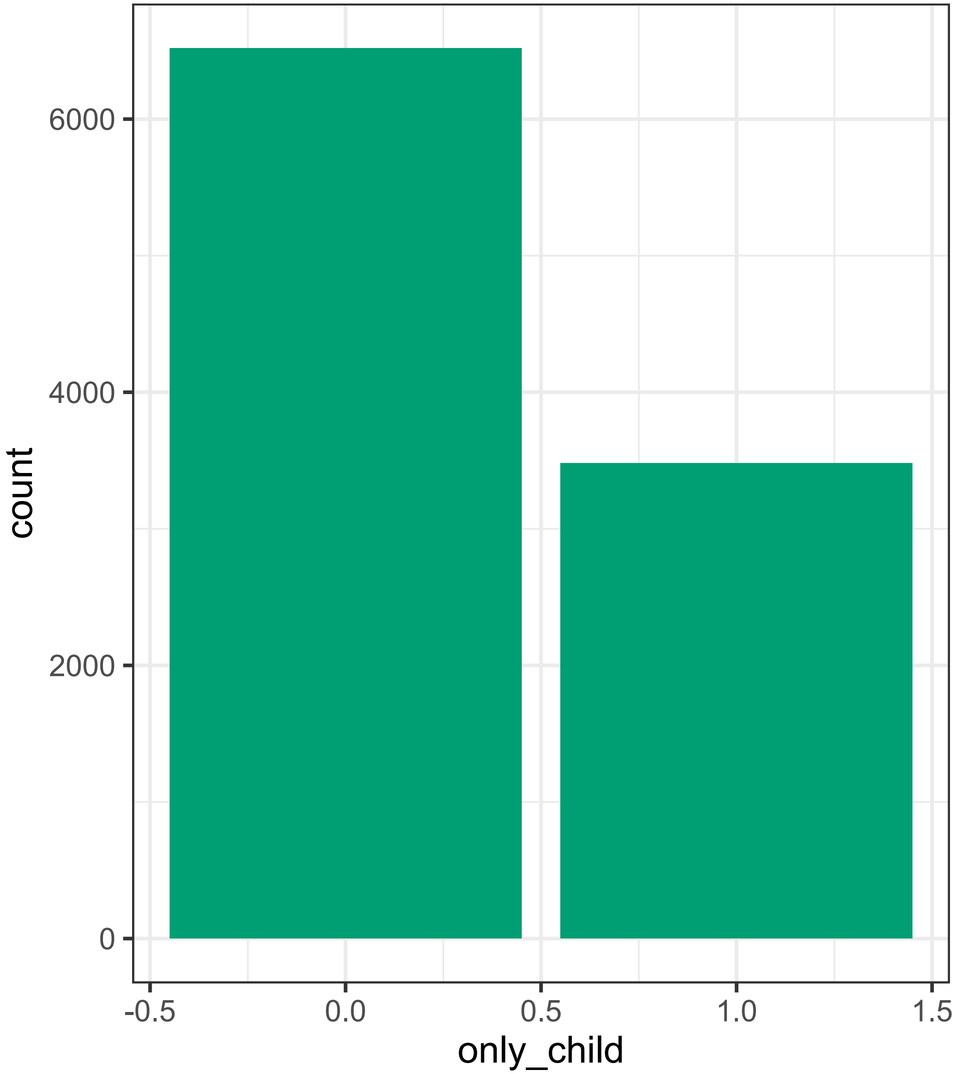
Basketball Team

```
ggplot(d, aes(x = basketball_team)) +
 geom_bar(fill = okabeito_colors(3)) +
 theme_bw()
```



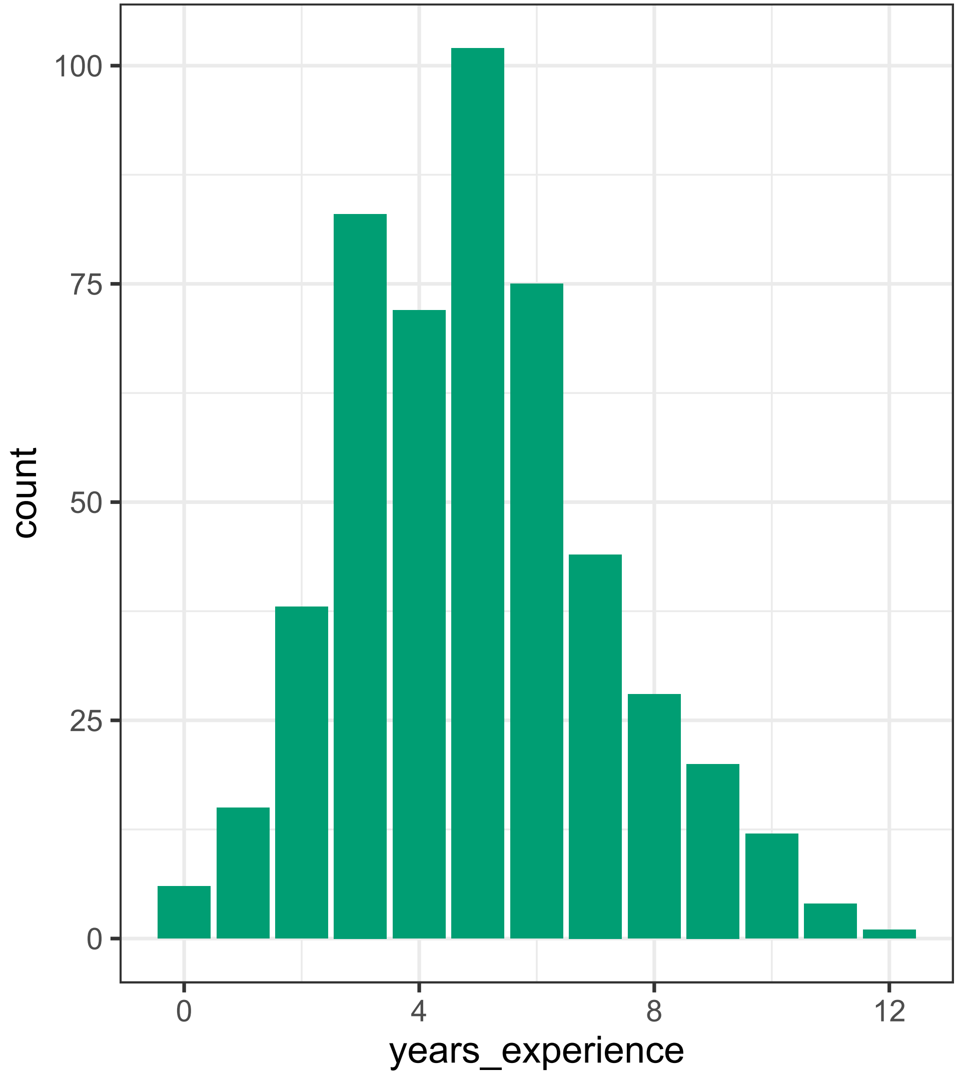
Only Child

```
ggplot(d, aes(x = only_child)) +
 geom_bar(fill = okabeito_colors(3)) +
 theme_bw()
```



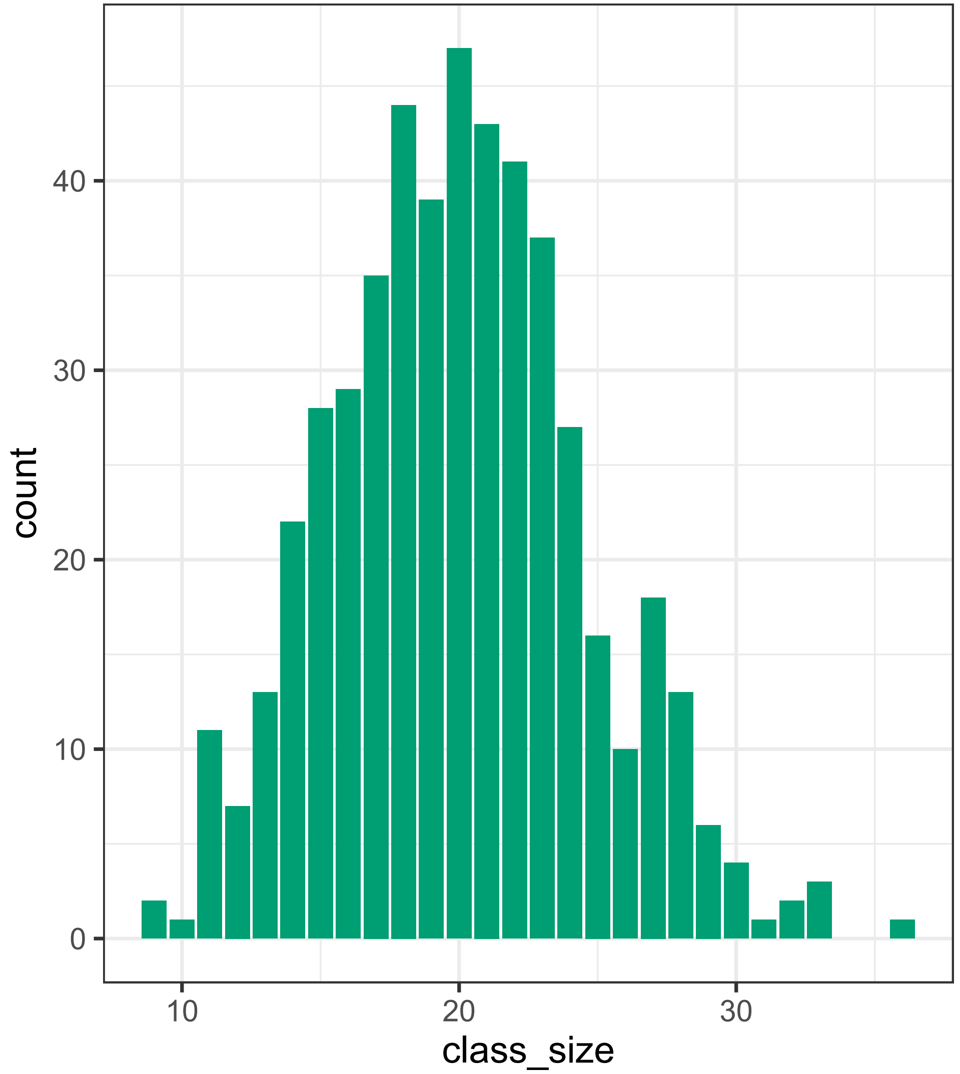
Years of Experience

```r
d |>
  select(teacher_id, years_experience) |>
  distinct() |>
  ggplot(aes(x = years_experience)) +
  geom_bar(fill = okabeito_colors(3)) +
  theme_bw()
```



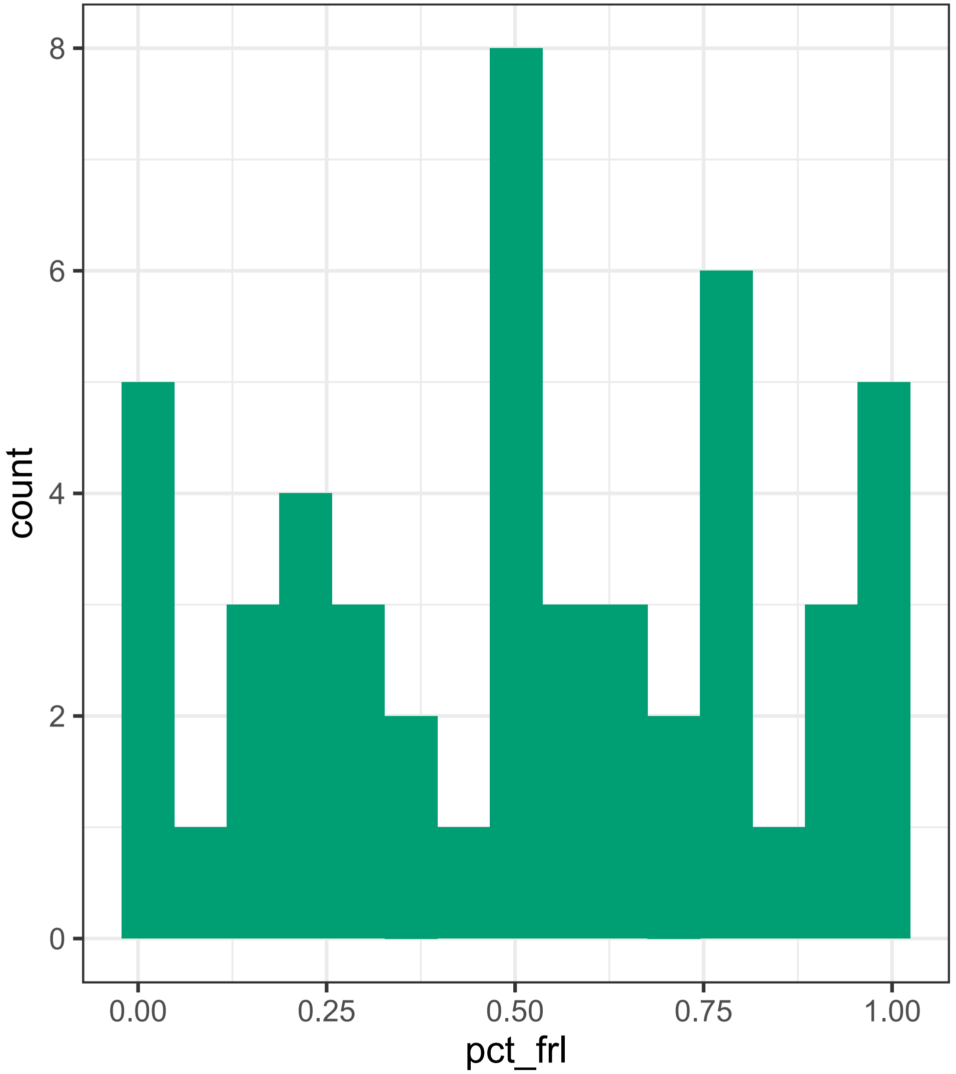
Class Size

```r
d |>
  select(teacher_id, class_size) |>
  distinct() |>
  ggplot(aes(x = class_size)) +
  geom_bar(fill = okabeito_colors(3)) +
  theme_bw()
```



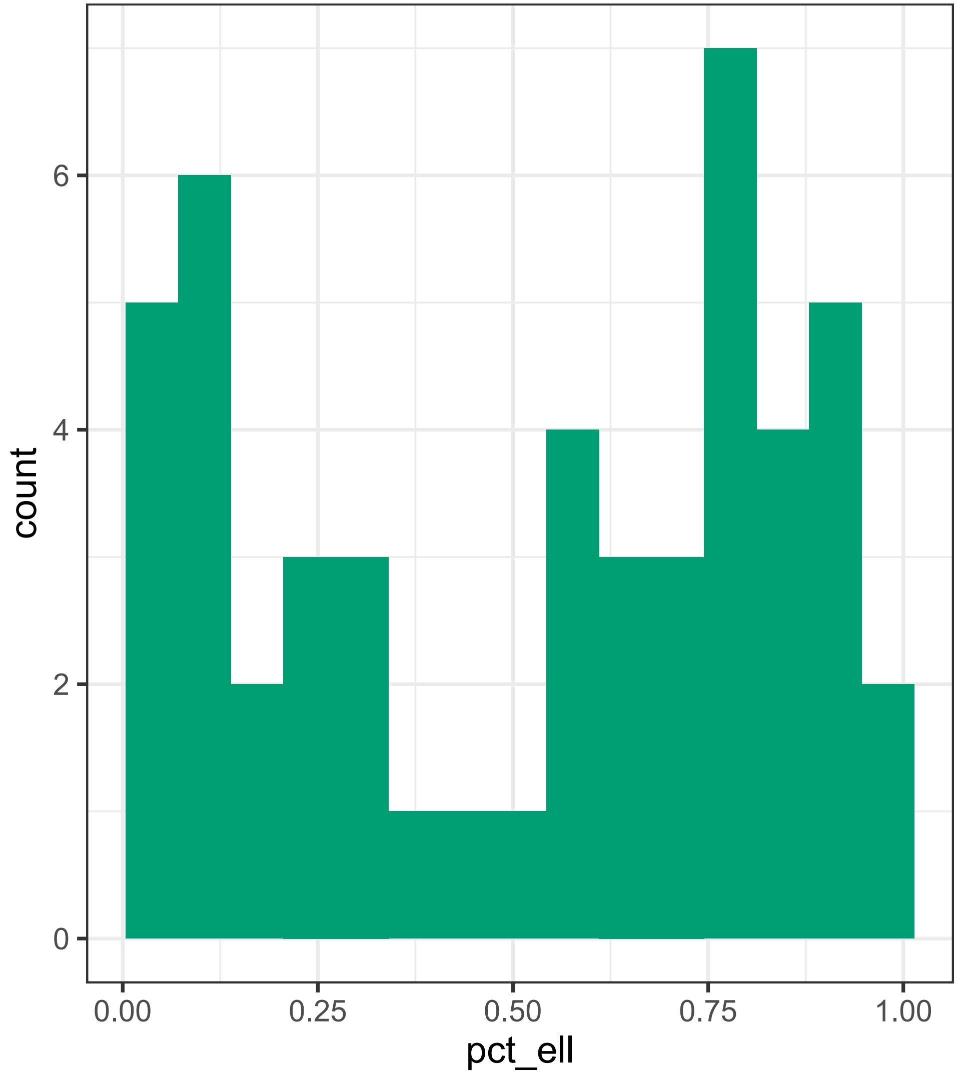
Percent Free/Reduced Price Lunch

```r
d |>
  select(school_id, pct_frl) |>
  distinct() |>
  ggplot(aes(x = pct_frl)) +
  geom_histogram(
    bins = 15,
    fill = okabeito_colors(3)) +
  theme_bw()
```



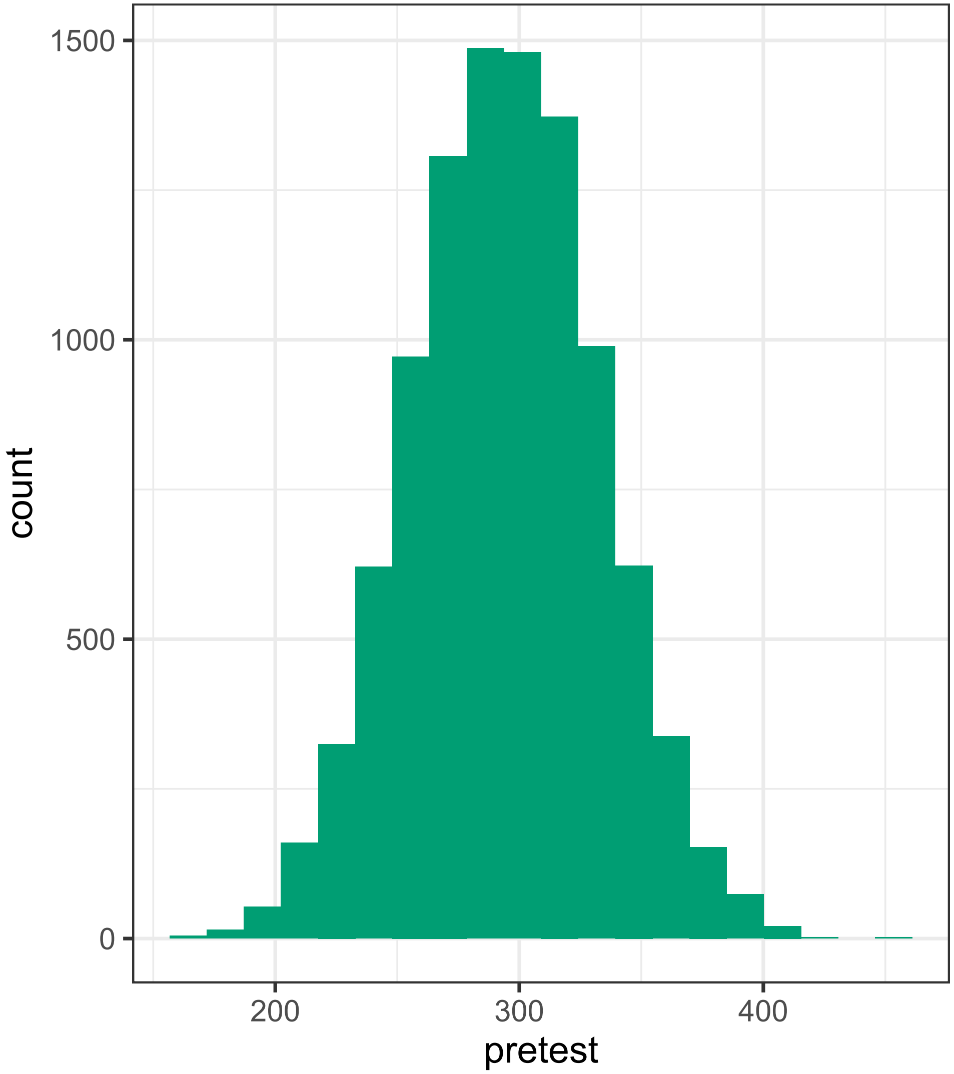
Percent English Language Learners

```r
d |>
  select(school_id, pct_ell) |>
  distinct() |>
  ggplot(aes(x = pct_ell)) +
  geom_histogram(
    bins = 15,
    fill = okabeito_colors(3)) +
  theme_bw()
```



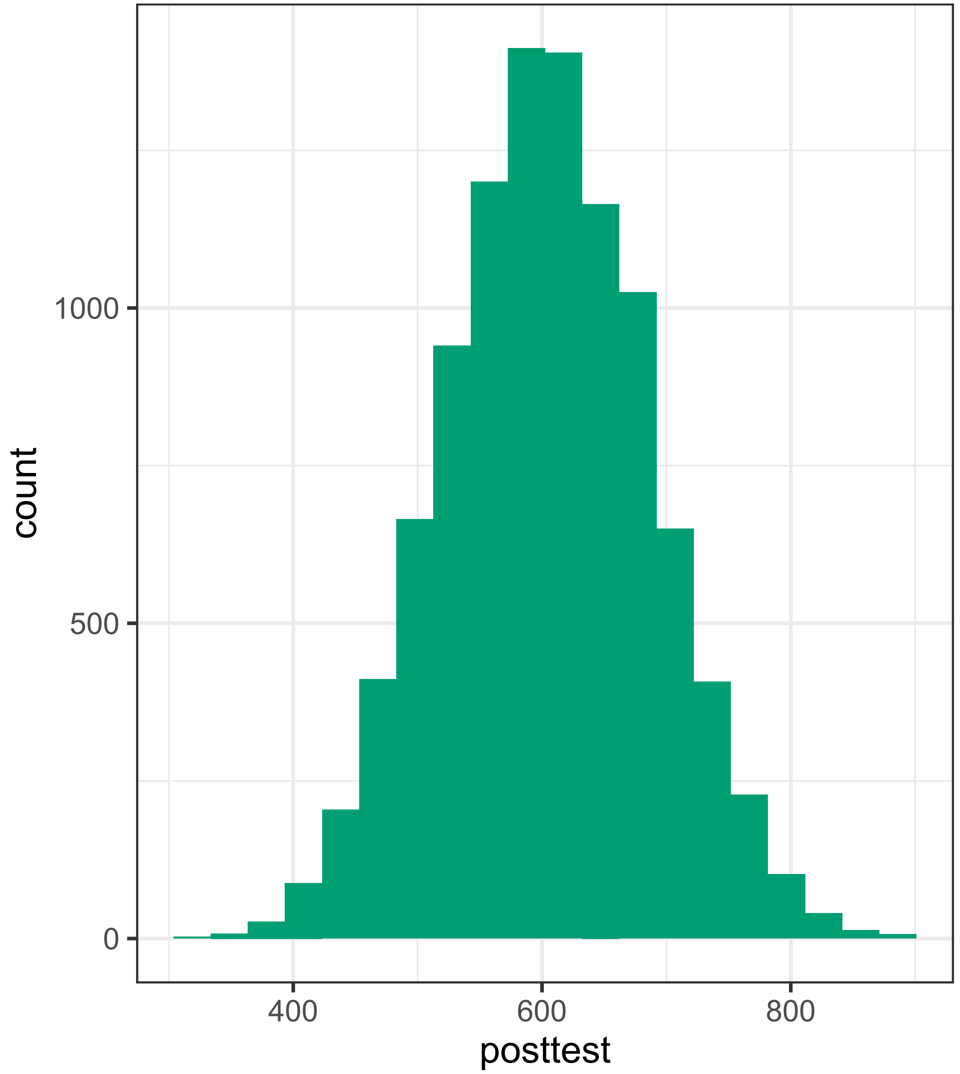
Pretest Score

```r
ggplot(d, aes(x = pretest)) +
  geom_histogram(
    bins = 20,
    fill = okabeito_colors(3)) +
  theme_bw()
```



Posttest Score

```r
ggplot(d, aes(x = posttest)) +
  geom_histogram(
    bins = 20,
    fill = okabeito_colors(3)) +
  theme_bw()
```



Thinking about Modeling

- For each variable (except the `_id` variables and `posttest`):
  1. If we include that variable in a regression with `posttest` as the outcome, how do we interpret the estimated coefficient?
  2. Can you include a random slope on that variable by teacher? If so, what does that mean? If not, why?
  3. Can you include a random slope on that variable by school? If so, what does that mean? If not, why?
- What does a random intercept by teacher do? How do you interpret that?
- What does a random intercept by school do? How do you interpret that?

Developing, Fitting, and Interpreting Multilevel Models

- We build multilevel models one step at a time
- Typically, start from a linear regression
- Add random effects one at a time and check model fit
- Adding some random effects will cause the model to not converge!
- Adding some other random effects won't improve the model at all
- Sometimes you may need to recheck effects you tried earlier after adding some other effects to the model
- One way to check for model fit is using `anova()`
- **I'll post my code after class, but you should attempt to follow along and take notes!**

Wrap Up

Recap

- Multilevel models help when observations are nested within groups or repeatedly measured
- In `R`, `lmer()` fits multilevel linear models and `glmer()` extends them to GLMs
- Try building MLMs one step at a time
- You know enough about MLMs to be dangerous; take Marc's course in the fall if you want to learn more!